SOKOINE UNIVERSITY OF AGRICULTURE

SCHOOL OF ENGINEERING AND TECHNOLOGY (SoET)

DEPARTMENT OF AGRICULTURAL ENGINEERING (DAE)

AE 316 MICROCOMPUTER SYSTEMS

2023-2024

SUPER ASSIGNMENT [Deadline: 26-Jan-2024]

1. Microcomputer, CPU and Memory units
   a) List a minimum of five microcontroller components (8 Marks)
   b) Provide a brief overview of three primary CPU components (6 Marks)
   c) Identify eight registers found in the Intel 8086 processor (8 Marks)
   d) Give a concise description of three primary programming levels (6 Marks)
   e) Explain briefly at least five features of PIC microcontrollers (5 Marks)

2. 68000 and PIC microprocessors
   a) Develop a program that begins at location 00000400H and adds the contents of memory addresses 00002020H and 00002030H. The sum of this operation, combined with the value 6482H, should be stored at address 00002050H. ***(5 Marks)***
   b) Modify the content at address 00002020H to 0B5FH and the content at address 00002030H to 38BFH. Afterward, perform the addition of these numbers and store the hexadecimal result at location 00002050H. ***(5 Marks)***
3. Arduino Programming
   a) Develop an Arduino program that showcases prime numbers, specifically those ending with the digit 3, and are less than 1000. ***(20 Marks)***
   b) Create an Arduino program that exhibits odd numbers below 400. Additionally, incorporate functionality to make the red LED blink when a number is divisible by 17. ***(20 Marks)***

c) Develop an Arduino program to manage two sets of traffic lights, each with three LEDs (red, green, and yellow). When one set displays red, the other set should show either green or yellow. Time intervals can be decided to prevent potential car collisions. ***(32 Marks)***

d) Create an Arduino program that allows the user to choose between displaying prime numbers and odd numbers below 200. Implement a menu system for user selection and output the chosen numbers accordingly. ***(10 Marks)***

e) Develop an Arduino program that calculates and prints the surface area and volume of a cylinder, considering user-provided values for the radius and height. Display the results with two decimal places. ***(10 Marks)***